Common and Domain-Specific Design Patterns for Mobile AR Applications to Visualize Measures on Building Facades in an Environmental Context

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Abstract: Mobile augmented reality (mAR) applications make it possible to visualize planned facade projects on-site with georeferencing and to scale. This paper presents two independently developed mAR prototypes and systematically compares them: Facade-Greening-AR for planning green facades (urban climate adaptation) and Facade-PV-AR for planning facade photovoltaics (energy transition). Both address the same physical domain, are based on an identical technical core (Unity, ARFoundation, Google Geospatial API, Google Streetscape Geometry API), and are aimed at citizens without specialized knowledge. The structured comparison across eight dimensions yields a catalog of seven design patterns: four cross-domain (georeferenced facade anchoring, visual in-situ assessment, local data persistence, domain-knowledge interface) and three domain-specific (discrete object placement, continuous area coverage, quantitative potential modeling). The catalog forms a generalizable foundation for future mAR applications in urban environmental contexts.

Keywords: Augmented Reality, green facades, facade photovoltaics, design patterns, requirements analysis, climate adaptation, renewable energy transition

1 Introduction

Cities must both reduce greenhouse gas emissions and adapt to the already unavoidable consequences of climate change [SM21]. Facade measures play a growing role in both areas of action: green facades, as nature-based solutions, mitigate urban heat islands, improve air quality, and enhance biodiversity [Pf16], while photovoltaics on facades contribute to decentralized electricity generation and thus to the decarbonization of the building sector [SS25].

A common challenge across both areas lies in communicating planned measures: stakeholders find it difficult to visualize what greening or PV systems would look like on an existing building and what ecological, economic, or aesthetic effects they would entail [St25, De26]. Mobile augmented reality (mAR) technology offers a promising solution: it enables a georeferenced, true-to-scale overlay of virtual objects onto the smartphone

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camera view directly on the facade on-site, making planned measures tangible [Az97].

mAR applications have been developed for various environmental planning tasks, such as visualizing urban greening measures [OSR23, Ci21], evaluating sites for renewable energy facilities [Ca16, Me23], or facilitating participation in planning processes [Bu22]. Although isolated design recommendations for urban AR planning exist, there has been no comparative analysis to date that distinguishes common patterns from domain-specific ones, thereby creating a transferable foundation for future developments.

This paper therefore addresses the following research question: What common and domain-specific design patterns can be identified for mAR applications used to visualize interventions on building facades in an environmental context? To answer it, two mAR prototypes developed in separate contexts at HTW Berlin are systematically compared: Facade-Greening-AR, an app for planning and visualizing facade greening [St25], and Facade-PV-AR, an app for planning and visualizing facade photovoltaics [De26]. The primary scientific contribution is a catalog of seven design patterns (M1 - M7) derived from two real-world implementations for mAR applications aimed at visualizing facade measures in an environmental context.

2 Background and Related Work

Green facades refer to the targeted greening of vertical building surfaces using soil- or wall-mounted planting systems [Pf23]. As a nature-based solution they block thermal radiation, cool the ambient temperature via evapotranspiration, filter pollutants and fine particulate matter from the air and contribute to biodiversity [Pf16], yet their adoption remains limited by planning barriers, a lack of imagination and insufficient information and advisory services [KKZ23]; assessing their potential quantitatively requires species-specific and seasonal parameters and is therefore difficult to standardize [THK19]. Facade PV, in turn, integrates solar modules into vertical building surfaces, either as curtain-wall facade elements or as building-integrated components (BIPV) [Hü18]. It converts incident solar radiation directly into electrical energy and can replace additional building materials [SM21], but is likewise held back by high investment costs, complex building code requirements and a lack of expertise among planners [SS25, Hü18]; its potential analysis covers technical, economic and environmental metrics, with orientation, tilt and shading as key influencing factors [SS25].

Augmented reality (AR) refers to the real-time overlay of physical reality with computer-generated, context-specific content [Az97]. For urban greening initiatives, mAR applications such as Cityscaper [Ci21] and GLARA [OSR23] visualize greening scenarios in street spaces, but do not address green facades. In building-integrated photovoltaics, AR has so far been used sporadically to highlight suitable facade surfaces [Ca16, Me23] and for shading simulation [VR15], while tools suitable for everyday use by laypeople are largely lacking [De26]. Comparative studies on design patterns in both domains are not yet available.

A georeferenced AR representation that accurately reflects the plan requires precise determination of the device's position [KH04], which GPS and compass typically cannot reliably achieve in urban environments [FFF16, St19]. Visual positioning services (VPS) offer significantly higher accuracy by aligning camera images with georeferenced reference images: the Google Geospatial API achieves position errors of approximately one meter and orientation errors of less than 5 degrees wherever Street View data is available [De26], and the complementary Streetscape Geometry API provides simple 3D meshes of surrounding buildings as a geometric basis for precise object placement on facades [St25, De26]. Both are part of the ARCore Extensions developed by Google.

3 Method: Comparative Design Study

To answer the research question, a comparative design study is conducted [SMM12]: two real-world system developments (Facade-Greening-AR and Facade-PV-AR) are systematically analyzed using a common comparative framework. The selection of systems follows the principle of maximum thematic coherence while ensuring maximum domain difference: both applications address the same physical object domain (vertical building facade), are based on a largely identical technical core, and target the same primary audience (citizens without specialized knowledge), yet operate in different environmental domains (climate adaptation vs. energy transition). This contrast enables the systematic separation of cross-domain design patterns from domain-specific ones.

The comparison is conducted across eight dimensions: requirements analysis (D1), georeferencing and facade detection (D2), object placement interaction (D3), object representation (D4), potential modeling (D5), data persistence (D6), UX and target audience suitability (D7), and domain-specific extensions (D8). The dimensions were inductively derived from the requirements documents of both prototypes, cover the entire development cycle from requirements gathering to UX, and were agreed upon in a workshop with all authors.

Both apps were developed within the same research group at HTW Berlin but in separate contexts: Facade-PV-AR by a research assistant as part of a research project [De26], Facade-Greening-AR as a bachelor's thesis [St25] supervised by that same research assistant. This ensures conceptual independence while sharing technical expertise, but does not completely rule out implicit assumptions arising from the supervisory relationship (see Section 7.2).

4 Facade-Greening-AR: Visualizing Green Facades

Domain, target audience and requirements. Facade-Greening-AR addresses urban climate adaptation through green facades. Its primary target audience consists of citizens without specialized knowledge who wish to explore the aesthetic potential of green facades on buildings; secondary audiences include municipalities, environmental organizations and professional planners. The requirements analysis is based on structured workshops with environmental agencies, municipalities and potential end users, and on a systematic review of related literature and comparable apps, whose gaps were formulated as negative requirements; requirements are assigned unique identifiers and prioritized (MUST/SHOULD/CAN). The core functional requirements are realistic AR visualization of greening objects on facades, interactive placement and editing of 3D plant objects, georeferenced anchoring and persistent design management (FR-1 to FR-4), while plant catalog, collaborative functions and climate data integration (FR-5 to FR-7) are optional.

Architecture, placement and rendering. The application is based on Unity 6 with ARFoundation and ARCore Extensions, utilizing the Google Geospatial API and the Streetscape Geometry API, and follows a three-tier architecture (presentation, logic, data); localization, facade recognition and anchor placement reside in the logic layer, and the data layer persists designs as local JSON files. Placement is performed with a simple tap: raycasting against the facade geometry provided by the Streetscape Geometry API determines the intersection point, where a georeferenced AR anchor is instantiated. Each anchor corresponds to a discrete plant object, scaled via a slider and rendered from a library of pre-built, static plant models (including ivy and wild grapevine) with lighting and shadows computed by Unity's AR rendering system. The geometric complexity of organic plant forms poses a particular challenge and requires level-of-detail (LOD) and billboard rendering techniques, which should be addressed in future work.

Potential modeling and implementation status. Potential modeling was deliberately not designed for Facade-Greening-AR: for the primary target audience visual appearance is the decisive factor, and the ecological potential of a green facade encompasses heterogeneous parameters that are difficult to standardize due to seasonal and species-specific influences, in contrast to the uniform energy yield of facade PV. The prototype has been implemented for iOS and Android with localization, facade detection, plant placement and local data persistence; the plant catalog, gesture-based scaling/rotation, cloud/backend synchronization and a share function have not yet been implemented.

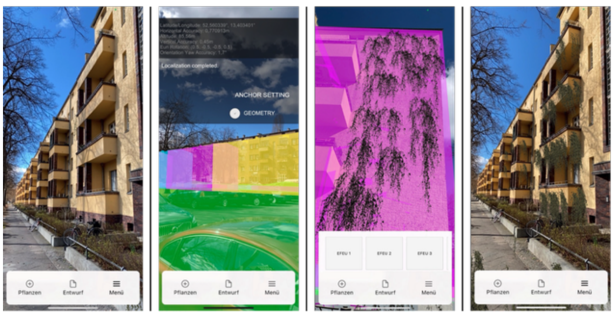

Fig. 1: mAR app Facade-Greening-AR (left: facade without AR greening; center-left: localization; center-right: plant selection; right: facade with AR greening)

5 Facade-PV-AR: Visualizing Facade Photovoltaics

Domain, target audience and requirements. Facade-PV-AR addresses the transition to renewable energy production using photovoltaics on building facades. Its primary target group consists of citizens who, without in-depth technical knowledge, wish to assess the visual appearance and energy potential of a facade PV system and thus generate a basis for deciding for or against its installation; secondary target groups include building owners, energy consultants and planning firms. The requirements were identified from the same two complementary sources and are likewise structured and prioritized [De26]. The MUST requirements are automatic capture of facade geometry and orientation-correct placement in the AR scene, interactive marking of sub-areas, automatic area-optimized module placement (conceptual; simplified in the prototype), manual adjustment of module type, color and spacing, and real-time calculation of technical, economic and ecological metrics (FR-01 to FR-05).

Architecture, placement and rendering. Facade-PV-AR uses the same technical core as Facade-Greening-AR: Unity 6, ARFoundation, Google Geospatial API and Streetscape Geometry API. Users mark the corner points of the area to be covered using tap gestures; the application then automatically determines an area-optimized layout from a catalog of real-world module types, with module spacing and type adjustable. The resulting layout is rendered in AR using static 3D models of the selected module type, placed automatically and rendered with standard shaders.

Potential modeling and implementation status. Every design change triggers an immediate recalculation of all key metrics: the energy potential of the facade is derived from the technical information of the panels, their orientation and the intensity of the sun's radiation at the given location, and from this, economic metrics (payback period, revenue) and environmental metrics (CO_2 savings compared to fossil fuel-based electricity generation) are derived and clearly presented. The prototype has been implemented for iOS and Android. A simplified version of the automatic layout generation lets the user mark a rectangle instead of a polygon, which is then textured with a PV module texture and displayed without real module type parameters; further implemented features are a rough facade potential calculation (approximation formula: 200 kWh/m^2 × factor × area of PV modules) and local storage of project files.

Fig. 2: mAR app Facade-PV-AR (left: mesh adjustment; center: facade marking; right: module and potential)

6 System Comparison and Design Patterns

6.1 Cross-Dimension Comparison

Dimension	Facade-Greening-AR	Facade-PV-AR
D1 Requirements analysis	Literature review and structured workshops; structured, prioritized requirements	Literature review and structured workshops; structured, prioritized requirements
D2 Georeferencing and facade detection	Geospatial API (VPS) + Streetscape Geometry API; raycasting via tap; manual calibration (transparent outlines)	Geospatial API (VPS) + Streetscape Geometry API; raycasting via crosshairs; manual calibration (transparent outlines)
D3 Object placement interaction	Location-specific: one tap creates one anchor per plant object	Area-continuous: corner points define an area that is filled with modules
D4 Object representation	Organic, complex meshes; many variants; high rendering effort; LOD/billboards; growth and seasonality	Rectangular, simple models; few variants; low rendering effort; standard shaders; static
D5 Potential modeling	None (deliberate design decision); aesthetic focus	Annual output, payback period, revenue, CO_2 savings; recalculated on every change; investment focus
D6 Data persistence	Local JSON (plant type, geolocation, rotation)	Local JSON (module coordinates, parameters, potential metrics)
D7 UX and target audience suitability	Citizens; usability a prioritized non-functional requirement; single-tap placement; no formal evaluation	Citizens; usability a prioritized non-functional requirement; marking of several points; no formal evaluation
D8 Domain-specific extensions	Plant catalog (location, care, ecology); optional climate and weather data	Module specifications (type, color, efficiency, dimensions)

Tab. 1: Comparison of Facade-Greening-AR and Facade-PV-AR across the eight dimensions D1 - D8

The two applications are largely congruent in D1, D2, D6, D7 and D8: they share the requirements process, the georeferencing stack, local JSON persistence, the primary target audience and the absence of formal usability testing. The substantive differences lie in D3, D4 and D5, and all three are functionally determined by the physical nature of the subject object rather than by design preference: a plant is location-specific, whereas PV modules cover a contiguous area (D3); organic plant geometry demands far more rendering effort and optimization than rectangular modules (D4); and only the investment-oriented PV case requires quantitative metrics alongside the visualization (D5). These three differences form the basis for M5, M6 and M7.

6.2 Derived Design Patterns M1 - M7

M1: Georeferenced facade anchoring (shared). *Context:* virtual objects must be anchored to real building facades precisely and persistently. *Solution:* a combination of VPS-based georeferencing (Google Geospatial API) and a facade geometry service (Google Streetscape Geometry API); raycasting against the facade geometry determines the exact placement point, and global coordinates from the Geospatial API persistently anchor it. *Evidence:* both systems use this pattern as their technical foundation.

M2: Visual in-situ assessment (shared). *Context:* non-experts should be able to assess the aesthetic impact of a facade measure without specialized software or static visualizations. *Solution:* AR overlay of virtual objects onto the real-time camera image on-site; the in-situ perspective under real lighting conditions creates an impact that static renderings cannot match. *Evidence:* both systems cite visual assessment of the building's appearance as their primary added value over existing planning tools.

M3: Local data persistence (shared, domain-specific expression). *Context:* planning drafts (anchor positions, object references, parameters) should be saved and restored between sessions. *Solution:* serialization of all planning-relevant data as human-readable JSON files in local storage. *Evidence:* both systems implement this pattern using an identical technical approach; only the JSON structure differs by domain.

M4: Domain knowledge interface (shared, domain-specific expression). *Context:* the placement decision depends on technical information that is not accessible to the primary target audience (plant suitability, module performance). *Solution:* structured linking of each placeable AR object to domain-specific technical information; shared as a pattern, domain-specific in its concrete form: plant catalog vs. module specifications. *Evidence:* both systems define such an interface.

M5: Discrete object placement (domain-specific: greening). *Context:* the object to be visualized is location-specific; each object exists at an individual, independent point on the facade. *Solution:* tap-based placement paradigm: a finger tap on the facade geometry generates a georeferenced anchor point for a single 3D object, intuitive and requiring no domain-specific prior knowledge. *Evidence:* Facade-Greening-AR; transferable to other location-specific facade objects (individual lights, information boards).

M6: Continuous area coverage (domain-specific: PV). *Context:* the individual objects to be visualized are discrete, but the facade area they cover is continuous and is to be occupied as a whole. *Solution:* area marking paradigm: users define the area to be covered by specifying corner points, and the application automatically fills it with the objects. *Evidence:* Facade-PV-AR; applicable to all facade measures covering a contiguous area (e.g. composite thermal insulation systems, flat solar thermal collectors).

M7: Quantitative potential modeling (domain-specific: PV). *Context:* the target group's decision is investment-oriented and requires quantitative metrics (yield, payback period, CO_2 savings). *Solution:* tight integration of AR visualization and real-time potential calculation; any change in the plan (area, orientation, module type) triggers an immediate, clearly presented recalculation of all relevant metrics. *Evidence:* Facade-PV-AR; applicable to other facade measures with measurable performance (thermal insulation: heating cost savings; green roofs: rainwater retention).

7 Discussion

7.1 Interpretation of the Design Patterns

The four common patterns (M1 - M4) form a stable, reusable foundation for future mAR applications for facade interventions: the combination of VPS-based georeferencing and Streetscape-Geometry-based facade geometry (M1) solves the fundamental technical problem of precise, persistent object anchoring to real building facades and, together with local data persistence (M3) and Unity/ARFoundation, forms a proven technical basis for an entire class of such applications. That this identical core supports fundamentally different interaction paradigms (M5 vs. M6) and information requirements (none vs. quantitative KPIs) underscores that technical convergence does not preclude conceptual divergence: the effort for domain-specific design (M4 - M7) is at least comparable to that for the shared technical foundation. For a new application it must therefore first be clarified whether the object is location-specific (tap) or area-continuous (area marking), and whether the primary decision is aesthetic (no metrics required) or investment-oriented (real-time potential calculation required); a structured link between the AR object and domain-specific information (M4) should be planned from the outset in either case.

7.2 Limitations

Lack of formal user studies: both prototypes were evaluated without formal user studies, so the derived patterns are based exclusively on requirements analyses and implementation decisions, not on empirical usage data; SUS-based usability tests with the primary target group are strongly recommended for further work. **Prototype nature:** both implementations are prototypes, and patterns M1 - M7 refer to the conceptual level, not the implementation level; this should be taken into account during interpretation. **Limited generalizability:** the pattern catalog is based on two case studies; additional mAR applications for other facade measures (e.g. thermal insulation or facade renovation) would strengthen the evidence base and potentially reveal further patterns. **Construct**

validity: the comparison dimensions were derived from the source documents of both systems, and independent validation by external experts is pending; since both systems were developed by the same team, implicit assumptions may have influenced the derivation of the patterns (confirmatory bias).

8 Summary and Outlook

This paper is the first to systematically compare two mobile AR applications for facade interventions in an environmental context (facade greening and facade photovoltaics). Seven design patterns (M1 - M7) were identified: four cross-domain (M1 - M4) and three domain-specific (M5 - M7). The common patterns demonstrate a stable, reusable technical and conceptual core for this class of applications: georeferenced facade anchoring via VPS and Streetscape Geometry, visual in-situ assessment as the primary added value, local data persistence, and a domain knowledge interface. The domain-specific patterns show that the physical nature of the subject object determines the placement paradigm (discrete vs. continuous) and that the target group's primary decision-making type (aesthetic vs. investment-oriented) determines the need for quantitative potential modeling.

The derived pattern catalog forms a generalizable basis for future mAR applications for facade interventions in an environmental context. Further work should empirically validate the patterns through formal user studies, deepen the requirements analysis for Facade-Greening-AR with more stakeholder surveys, and expand the catalog with case studies from related facade domains (e.g. thermal insulation, renovation).

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References

[Az97] Azuma, R. T.: A Survey of Augmented Reality. Presence 6(4), pp. 355 - 385, 1997.

[Bu22] Burkard, S. et al.: A Mobile Augmented Reality Application for Displaying Planned Wind Turbines. In: Environmental Information Systems. Springer, Wiesbaden, pp. 21 - 41, 2022.

[Ca16] Carmo, M. B. et al.: Augmented Reality for Decision Support on Solar Radiation Harnessing. In: 23rd Portuguese Meeting on Computer Graphics and Interaction (EPCGI). IEEE, pp. 1 - 8, 2016.

[Ci21] Cityscaper GmbH: CITYSCAPER - Digital Citizen Participation with AR in Urban

Planning. https://page.cityscaper.de, 2021.

[De26] Deharde, M. et al.: Mobile AR Application for the Integrated Visualization and Potential Assessment of Building-Integrated Photovoltaics. In: Environmental Information Systems. Springer, Wiesbaden, 2026 (forthcoming).

[FFF16] Fedorov, R.; Frajberg, D.; Fraternali, P.: A Framework for Outdoor Mobile Augmented Reality and Its Application to Mountain Peak Detection. In: AVR 2016, LNCS 9768. Springer, Cham, 2016.

[Hü18] Hülsmeier, F. et al.: SOLAR.shell. Fraunhofer IRB Verlag, 2018.

[KH04] Karimi, H. A.; Hammad, A. (eds.): Telegeoinformatics. CRC Press, Boca Raton, 2004.

[KKZ23] Knifka, W.; Karutz, R.; Zozmann, H.: Barriers and Solutions to Green Facade Implementation. Buildings 13(7), 1621, 2023.

[Me23] Meireles, C. et al.: Mobile Augmented Reality for Analysis of Solar Radiation on Facades. In: Proc. VISIGRAPP 2023, Vol. 1: GRAPP. SCITEPRESS, pp. 131 - 142, 2023.

[OSR23] Othengrafen, F.; Sievers, L.; Reinecke, E.: Using Augmented Reality in Urban Planning Processes. GAIA 32(1), pp. 54 - 63, 2023.

[Pf16] Pfoser, N.: Fassade und Pflanze. Dissertation, TU Darmstadt, 2016.

[Pf23] Pfoser, N.: Grüne Fassaden. DETAIL Praxis, Munich, 2023.

[SM21] Schröder, J.; Moebus, S.: Climate-Sensitive Urban Planning and Urban Development. In: Versorgungs-Report Climate and Health. MWV, 2021.

[SMM12] Sedlmair, M.; Meyer, M.; Munzner, T.: Design Study Methodology: Reflections from the Trenches and the Stacks. IEEE TVCG 18(12), pp. 2431 - 2440, 2012.

[SS25] Schram, W. L.; Shirazi, E.: PV on Facades: A Financial, Technical, and Environmental Assessment. Energy and Buildings 328, 2025.

[St19] Stranner, M. et al.: A High-Precision Localization Device for Outdoor AR. In: ISMAR-Adjunct. IEEE, pp. 37 - 41, 2019.

[St25] Starzl, D.: Digital Design of Greening Measures: Concept and Prototype Implementation of a Mobile AR App. Bachelor’s thesis, HTW Berlin, 2025.

[THK19] Tudiwer, D.; Hollands, J.; Korjenic, A.: Berechnung der Kühlgestehungskosten von fassadengebundenen Begrünungssystemen im städtischen Raum. Bauphysik 41, pp. 120 - 124, 2019.

[VR15] Veldhuis, A. J.; Reinders, A. H. M. E.: Shadow Analysis for BIPV and PIPV Systems in a Virtual Environment. In: 42nd IEEE PVSC, pp. 1 - 5, 2015.